of 1.9A). So, (i) is implied by the Lebesgue theorem on dominated convergence; (ii) is obvious; to prove (iii), note that

$$\|\widehat{P}_\Sigma U(\tau)h\|^2 = \int_\Sigma |\widehat{h}(\xi-\tau)|^2\,d\tau = \int_{\Sigma+\tau} |\widehat{h}(\xi)|^2\,d\xi.$$

The last integral tends to zero as $|\tau| \to +\infty$. Now, P_S and $U(\tau)$ commute for every $\tau \in \mathbb{R}^d$, and the flexibility of $(P_S, \widehat{P}_\Sigma)$ is proved. The flexibility of $(\widehat{P}_\Sigma, P_S)$ can be proved analogously. $\square$

Hence $(P_S, \widehat{P}_\Sigma)$ satisfies (11) and (12) of §1 (its compactness was proved in Sect. 3).

C) To answer the question stated in A) we only have to reproduce the result of 1.6C) for $P := P_S$, $Q := \widehat{P}_\Sigma$ which becomes *the Slepian–Pollak theorem*. It is one more form of the UP and at the same time a quantitative refinement of the Amrein–Berthier theorem which forbids the point $(\|Ph\|, \|Qh\|) \in \mathbb{R}^2$ to coincide with the vertex $(1,1)$ of the square $J \times J$ ($P = P_S$, $Q = \widehat{P}_\Sigma$, $h \in L^2$,, $\|h\| = 1$). Fig. 1.2 shows that if (5) is satisfied, then this point must be *bounded off* $(1,1)$, and $\|QPQ\|(< 1)$ determines how far (the zone allowed for $(\|Ph\|, \|Qh\|)$ is marked on Fig. 1.2). Under conditions stated in B) the (open) domain bounded by segments $J := [0,1]$, $[0,i]$, $[1, 1+ic]$, $[i, c+i]$ and by the curve $\gamma := \{(x,y) : \arccos x + \arccos y = \arccos c\}$ coincides with the interior of $\mathcal{W}\,(= \mathcal{W}_{P,Q})$, according to (18) of §1. A more subtle question is whether $\mathcal{W}$ is closed. Under the same conditions the origin always belongs to $\mathcal{W}$ (see 5B)). As to the vertices $p := (0,1)$ and $q := (1,0)$ of the square $J \times J$, they may belong to $\mathcal{W}$ and may not belong to it (see the Kargayev theorem in §3.9 of Part Two). They are surely not in $\mathcal{W}$ if S and Σ are bounded (or if $d = 1$, and S and Σ are semibounded); this follows from the F. and M. Riesz theorem. The condition $p \notin \mathcal{W}$, $q \notin \mathcal{W}$ is equivalent with (17) of §1, and then both $[0,1)$ and $[0,i)$ are in $\mathcal{W}$ (see concluding remarks in 1.6H)). If $p \in \mathcal{W}$ (or if $q \in \mathcal{W}$), then, by 1.6F), $[0,1) \subset \mathcal{W}$ (resp., $[0,i) \in \mathcal{W}$).

§3. Functions with Sparse Spectra. The Mikheyev Theorem

Consider a Lebesgue measurable set $E \subset \mathbb{T}$ and a set $\Lambda \subset \mathbb{Z}$. We say that (E, Λ) is a strong a-pair if there is a number $c = c(E, \Lambda)$ such that

$$\|f\|_2^2 \le c\Big(\int_{E'} |f|^2 + \sum_{n\in\Lambda'} |\widehat{f}(n)|^2\Big) \text{ for any } f \in L^2(\mathbb{T}) \tag{1}$$

where $E' := \mathbb{T} \setminus E$, $\Lambda' := \mathbb{Z} \setminus \Lambda$ (see the estimate (2) in the Introduction). We say that Λ is *sparse* (and write $\Lambda \in$ SPARSE) if (1) holds for any E with

$m(E) < 1$. The principal result of this paragraph is the Mikheyev Theorem on the relation of sparse sets with classes $\Lambda(2)$ and $\Lambda^*(2)$ (to be defined in Sections 3 and 4). The Mikheyev theorem is stated in Sect. 4 and proved in Sect. 5–7. In Sect. 8 we discuss the Hadamard lacunary sets.

1. Remarks on the Strong Annihilation.
Suppose $\Lambda \subset \mathbb{Z}$, $A \subset \mathbb{T}$, $\gamma > 0$. If

$$f \in L^2(\Lambda) \implies \int_A |f|^2 \geq \gamma \|f\|_2^2 \tag{2}$$

then we say that Λ is *γ-sufficient for A*. (The symbol $L^2(\Lambda)$ denotes $\{f \in L^2(\mathbb{T}) : \operatorname{spec} f \subset \Lambda\}$). Recall that (A', Λ) is a strong a-pair iff Λ is γ-sufficient for A with a positive γ (see §1).

A) A sparse set Λ is uniformly sufficient for all sets of length close to one. To be more precise, *there are $\gamma > 0$, $\delta > 0$ such that* (2) *holds for any A satisfying* $|A| \geq 1 - \delta$ (we write $|A|$ instead of $m(A)$).

Proof. If no such pair γ, δ exists, then there are sequences (A_k), (f_k) such that $|A'_k| \leq 2^{-(k+1)}$, $\int_{A_k} |f_k|^2 \leq k^{-1} \|f_k\|_2^2$. But then $(\Lambda, \cup A'_k)$ is not a strong a-pair, though $|\cup A'_k| \leq 1/2$. □

B) Taking A' instead of A, we may rephrase A) as follows: if Λ is sparse, then there are numbers $\delta > 0$, $\gamma \in (0,1)$ such that

$$|B| < \delta,\ f \in L^2(\Lambda) \implies \int_B |f|^2 \leq (1-\gamma)^2 \|f\|_2^2.$$

C) Let δ, γ be positive numbers. We say that *the pair (E, Λ) has the (δ, γ)-property if*

$$E_1 \subset E,\ |E \setminus E_1| \leq \delta,\ f \in L^2(\Lambda) \implies \int_{E_1} |f|^2 \geq \gamma \|f\|_2^2.$$

The same argument as in A) shows that *if Λ is sparse, then (E, Λ) has a (δ, γ)-property whenever* $|E| > 0$.

D) Suppose $\widetilde{E} \supset E$, and (E, Λ) has the (δ, γ)-property. Then so does $(\widetilde{E}, \Lambda)$. Indeed, if $E_1 \subset \widetilde{E}$, then $\widetilde{E} \setminus E_1 \supset E \setminus E_1 = E \setminus (E \cap E_1)$. Hence $|E \setminus (E \cap E_1)| \leq \delta$ whenever $|\widetilde{E} \setminus E_1| \leq \delta$.

2. The Union of a Sparse Set and of a Finite Set is Sparse. This fact is implied by 1C) and by the following assertion. *Suppose (E, Λ) has (δ, γ)-property, $\ell \in \mathbb{Z}$, $\Lambda^* := \Lambda \cup \{\ell\}$. Put $\gamma' := |E|(\gamma/10)^{(\delta+1)/\delta}$. Then Λ^* is γ'-sufficient for E.*

Proof. Suppose $f \in L^2(\Lambda^*)$, $\xi \in \mathbb{T}$, $|\xi E \setminus E| \leq \delta$. Put $\tau^2 := \int_E |f|^2$, $g(z) := f(z) - \xi^\ell f(\bar{\xi} z)$. Then $\widehat{g}(\ell) = 0$, and $g \in L^2(\Lambda)$. But $|E \setminus (E \cap \xi E)| = |\xi E \setminus E| \leq \delta$, whence

$$2\tau = \tau + \left(\int_{\xi E} |f(\bar{\xi}z)|^2\, dm(z)\right)^{1/2} \geq \left(\int_{E\cap\xi E} |f(z) - \xi^\ell f(\bar{\xi}z)|^2\, dm(z)\right)^{1/2} =$$

$$= \left(\int_{E\cap\xi E} |g|^2\right)^{1/2} \geq \sqrt{\gamma}\|g\|_2 \geq \sqrt{\gamma}\left(\int_{\xi E\setminus E} |g|^2\right)^{1/2} \geq$$

$$\geq \sqrt{\gamma}\left[\left(\int_{\xi E\setminus E} |f|^2\right)^{1/2} - \tau\right].$$

Thus $\int_{\xi E\setminus E} |f|^2 \leq (1 + 2/\sqrt{\gamma})^2\tau^2$, and

$$\int_{E\cup\xi E} |f|^2 \leq ((1 + 2/\sqrt{\gamma})^2 + 1)\tau^2 \leq (10/\gamma)\tau^2,$$

since $\gamma < 1$. Suppose there is a $\xi \in \mathbb{T}$ such that $|\xi E \setminus E| = \delta$, and put $E^1 := E \cup \xi E$. Then $E^1 \supset E$, $|E^1| \geq |E| + \delta$, $\int_{E^1} |f|^2 \leq 10\gamma^{-1}\tau^2$ for any $f \in L^2(\Lambda^*)$. But (E^1, Λ) has the (γ, δ)-property (see 1D)). If $|\xi E^1 \setminus E_1| = \delta$ for a $\xi \in \mathbb{T}$, then we may construct a set $E^2 \supset E^1$ such that $|E^2| \geq |E^1| + \delta \geq$ $\geq |E| + 2\delta$, $\int_{E^2} |f|^2 \leq 10\gamma^{-1} \int_{E^1} |f|^2 \leq 100\gamma^{-2}\tau^2$ for any $f \in L^2(\Lambda^*)$. We may repeat this procedure no more than $k := [1/\delta]$ times (otherwise we get a subset E_k of $\mathbb{T}$ with $|E_k| > 1$). This means that no point $\xi \in \mathbb{T}$ satisfies $|\xi E_k \setminus E_k| \geq \delta$. Hence

$$\int_{\xi E_k} |f|^2 \leq (10/\gamma)^{k+1} \int_E |f|^2 \text{ for any } f \in L^2(\Lambda^*) \text{ and for any } \xi \in \mathbb{T}.$$

Integrating over $\mathbb{T}$, we get $|E_k|\, \|f\|_2^2 \leq (10\gamma^{-1})^{k+1}\tau^2$ ($f \in L^2(\Lambda^*)$). But $|E| \leq$ $\leq |E_k|$, $k \leq 1/\delta$. Thus Λ^* is γ'-sufficient for E. □

3. Classes $\Lambda(p)$. We denote by $\mathcal{P}(\Lambda)$ the set of all Λ-polynomials (i.e., $\{ f \in C(\mathbb{T}) : \operatorname{spec} f \text{ is a finite subset of } \Lambda \}$).

A) Fix a positive number p. *The following assertions are equivalent*:

(a) *there are* $\delta > 0$, $c > 0$ *such that*

$$E \subset \mathbb{T},\ |E| < \delta,\ f \in \mathcal{P}(\Lambda) \implies \int_E |f|^p \leq (1 - c)\|f\|_p^p;$$

(b) *there are* $q \in (0, p)$, $C > 0$ *such that*

$$f \in \mathcal{P}(\Lambda) \implies \|f\|_p \leq C\|f\|_q;$$

(c) *for every* $q \in (0, p)$ *there is a* $C(q)$ *such that*

$$f \in \mathcal{P}(\Lambda) \implies \|f\|_p \leq C(q)\|f\|_q.$$

Proof. (c)$\Rightarrow$(b) being obvious, let us prove (b)$\Rightarrow$(a): $C^{-q}\|f\|_p^q \le \|f\|_q^q = \int_E |f|^q + \int_{E'} |f|^q \le |E|^{1-q/p}\|f\|_p^q + (\int_{E'} |f|^p)^{q/p}$; if $|E| < \delta$ and $\delta^{1-q/p} < 1/2C^q$, then $(\int_{E'} |f|^p)^{q/p} \le \|f\|_p^q/2C^q$, whence $\int_{E'} |f|^p \ge c\|f\|_p^p$ and $\int_E |f|^p \le (1-c)\|f\|_p^p$ $(c := 2^{p/q}C^p)$.
To prove (a)$\Rightarrow$(c), put $E := \{\zeta \in \mathbb{T} : |f(\zeta)| > \delta^{-1/q}\|f\|_q\}$ and note that $|E| < \delta$. Hence $\|f\|_p^p \le c^{-1}\int_{E'} |f|^p \le c^{-1}\delta^{-1/q}\|f\|_p^{q-p}\int_{E'} |f|^q \le c^{-1}\delta^{-1/q}\|f\|_q^p$, and $C_q := c^{-1/p}\delta^{-1/pq}$ enjoys the desired property. □

B) The class of sets Λ satisfying (one of) the conditions (a)–(c) is denoted by $\Lambda(p)$. Clearly, $\Lambda(p) \subset \Lambda(q)$ if $p > q$ (this follows from (c)). The inclusion $\Lambda \in \Lambda(p)$ is a very indirect way to say that the set Λ is "rarefied". A more explicit characterization of sparseness of $\Lambda(p)$-sets will be given in D) and E). The result of Sect. 1 means that SPARSE $\subset \Lambda(2)$.

C) Suppose $n \in \mathbb{Z}$, $m_1, \dots, m_k \in \mathbb{N}$. The set

$$I(n; m_1, \dots, m_k) := \{n + \sum_{j=1}^{k} \varepsilon_j m_j : \varepsilon_j = 0 \text{ or } 1\}$$

is called *a cluster* of length k (corresponding to parameters $n, m_1, \dots, m_k$). The same cluster can correspond to different parameters and to different k. For instance,

$$I(0; \underbrace{1, 1, \dots, 1}_{2^{\ell-1}}) = I(0; 1, 2, 4, \dots, 2^{\ell-1}).$$

To avoid this difficulty, we impose a sparseness condition on $m_1, \dots, m_k$ (as we did in §2.3). Namely, let us say that a cluster is *regular* if $m_{j+1} \ge 5m_j$, $j = 1, \dots, k-1$. *Any cluster of length* 6^k *contains a regular cluster of length* k.

Proof. We may assume that $m_1 \le \dots \le m_{6^k}$, since a cluster does not depend on the order of m_j. But then $J := I(n; \sum_{j=1}^{6} m_j, \sum_{j=6+1}^{6^2} m_j, \dots, \sum_{j=6^{k-1}+1}^{6^k} m_j)$ is regular, and $J \subset I(n; m_1, \dots, m_{6^k})$. □

D) Suppose $I = I(n; m_1, \dots, m_k)$ is a regular cluster. Consider the Riesz product $f_t := \prod_{j=1}^{k}(1 + tz^{m_j})$ with a small positive parameter t (see §2.3). It is easy to see that $\operatorname{spec} f_t \subset I$. Let us estimate $\|f_t\|_p$ $(p > 0)$:

$$\begin{aligned}|1 + tz^{m_j}|^p =& (1 + tz^{m_j})^{p/2}(1 + t\bar{z}^{m_j})^{p/2} = \\ =& (1 + \tfrac{p^2}{4}t^2) + \tfrac{pt}{2}(z^{m_j} + \bar{z}^{m_j}) + \tfrac{p}{2}(\tfrac{p}{2} - 1)(z^{2m_j} + \bar{z}^{2m_j})t^2 + \\ &+ O(t^3) \qquad (t \to +0,\ z \in \mathbb{T}).\end{aligned}$$

Multiplying these estimates, we get

$$\prod_{j=1}^{k}\left[(1+\tfrac{p^2t^2}{4}-C_pt^3)+\tfrac{pt}{2}(z^{m_j}+\bar z^{m_j})+\tfrac{p}{2}(\tfrac{p}{2}-1)(z^{2m_j}+\bar z^{2m_j})t^2\right]\le|f_t(z)|^p\le$$

$$\le\prod_{j=1}^{k}\left[(1+\tfrac{p^2t^2}{4}+C_pt^3)+\tfrac{pt}{2}(z^{m_j}+\bar z^{m_j})+\tfrac{pt^2}{2}(\tfrac{p}{2}-1)(z^{2m_j}+\bar z^{2m_j})t^2\right]\quad(3)$$

where C_p is a positive constant depending only on p. Both products can be represented as $(1+p^2t^2/4\pm C_pt^3)+Q$ where Q is a polynomial, and $0\notin\operatorname{spec}Q$ (since $m_{j+1}\ge 5m_j$). Integrating (3) over $\mathbb{T}$, we conclude that

$$(1+(p^2/4)t^2-C_pt^3)^k\le\int_{\mathbb{T}}|f_t|^p\le(1+(p^2/4)t^2+C_pt^3)^k$$

whence $(1+(p/4)t^2-C_p't^3)^k\le\|f_t\|_p\le(1+(p/4)t^2+C_p't^3)^k$. Suppose $q<p$ and pick a small $t\in(0,1)$ satisfying $1+(q/4)t^2+C_qt^3<1+(p/4)t^2-C_pt^3$. If k is sufficiently large, then the ratio $\|f_t\|_q/\|f_t\|_p$ is arbitrarily small. Summing up this argument we may say that if $\Lambda\in\Lambda(p)$, then Λ does not contain too long regular clusters. Thus, according to C), it does not contain too long clusters. In other words, *there is a number majorizing the length of any cluster contained in* Λ. As we shall see in the next subsection, this property implies a serious restriction concerning the density of Λ on large segments of $\mathbb{Z}$.

E) *If Λ contains no cluster of length ℓ, then no more than $C_\ell N^{(1-2^{1-\ell})}=:$ $=:D(\ell,n)$ of any N successive integers belong to Λ.*

Proof. We proceed by induction (with respect to ℓ). Any couple of (different) integers is a cluster of length one. Hence our assertion is trivial for $\ell=1$. Suppose it is true for an ℓ, and Λ contains no cluster of length $\ell+1$. Put $\Lambda_j:=$ $:=\{k\in\Lambda:\ k+j\in\Lambda\}$, $j=1,2,\ldots$. If Λ_j contains a cluster $I(n;m_1,\ldots,m_k)$ of length k, then Λ contains the cluster $I(n;m_1,\ldots,m_k;j)$ of length $k+1$. Hence Λ_j contains no cluster of length ℓ. Take a positive integer s and note that $\Lambda':=\Lambda\setminus\bigcup_{j=1}^{s}\Lambda_j$ contains no pair (λ_1,λ_2) with $j:=\lambda_2-\lambda_1=s$ (otherwise $\lambda_1\in\Lambda_j$). Thus no more than N/s of any N successive integers belong to Λ'; according to our assumption, no more than $D(\ell,n)$ of them are in Λ_j (for any $j=1,2,\ldots$). Thus no more than $sD+Ns^{-1}$ of N successive integers are in Λ. Taking $s:=[N^{2^{-\ell}}]$ we get the desired result. □

4. Classes $\Lambda^*(p)$. Statement of the Main Result. A set $\Lambda\subset\mathbb{Z}$ is said to belong to $\Lambda^*(p)$ $(p>0)$ if for any $\varepsilon>0$ there is a $\delta>0$ such that $\int_E|f|^p\,dm\le\varepsilon\|f\|_p^p$ for any $E\subset\mathbb{T}$ with $|E|<\delta$ and for any $f\in\mathcal{P}(\Lambda)$. In other words, $\Lambda\in\Lambda^*(p)$ means that the set functions $E\mapsto\int_E|f|^p\,dm$ where $f\in\mathcal{P}(\Lambda)$ and $\|f\|_p\le1$ are uniformly absolutely continuous with respect to m. The inclusion $\Lambda^*(p)\subset\Lambda(p)$ is obvious (see A)). On the other hand, if $q>p$, $\Lambda\in\Lambda(q)$ and $f\in\mathcal{P}(\Lambda)$, then $\int_E|f|^p\le|E|^{1-p/q}\|f\|_q^p\le|E|^{1-p/q}\|f\|_p^p$. Hence

$$\Lambda(p+0) := \underset{q>p}{\cup} \Lambda(q) \subset \Lambda^*(p) \subset \Lambda(p).$$

It is known that $\Lambda(p+0) = \Lambda(p)$ ($= \Lambda^*(p)$) for $p < 2$. J. Bourgain has shown that $\Lambda(p) \setminus \Lambda^*(p) \neq \emptyset$ for $p > 2$. For $p = 2$ the situation remains unclear. Unfortunately, just this particular case is the most interesting for us.

The Mikheyev Theorem. $\Lambda^*(2) \subset \mathrm{SPARSE} \subset \Lambda(2)$

The right inclusion has been already proved (see A), B)). The proof of the left one is given in Sect. 7. In Sect. 8 we give a very simple sufficient condition for a set to belong to $\Lambda^*(2)$. Note that the union of two $\Lambda^*(2)$-sets is again in $\Lambda^*(2)$.

5. Almost Orthogonal Splitting of a Function. A set $\Lambda_0 \in \Lambda^*(2)$ and a Lebesgue measurable set $E \subset \mathbb{T}$ of positive length will remain fixed throughout the proof. Our aim is to show that (2) holds for $\Lambda := \Lambda_0$ and $A := E$ (with *a positive* γ).

A) Let N be a positive integer. We say that the sets Λ', $\Lambda'' \subset \mathbb{Z}$ are *N-separated* if $|\ell' - \ell''| > N$ for any $(\ell', \ell'') \in \Lambda' \times \Lambda''$.

Suppose $\Lambda \subset \Lambda_0$, $\Lambda = \cup \Lambda_j$ and Λ_j are disjoint. Then any $f \in L^2(\Lambda)$ can be written as $\sum f_j$, $\operatorname{spec} f_j \subset \Lambda_j$. This decomposition is orthogonal in $L^2(\mathbb{T})$. We are going to show it is *almost* orthogonal in $L^2(E)$ if Λ_j are N-separated, and N is sufficiently large. *For any $\varepsilon > 0$ there is an $N = N(\varepsilon) \in \mathbb{N}$ such that*

$$f \in L^2(\Lambda) \implies \triangle := \left| \int_E |f|^2 - \sum_j \int_E |f_j|^2 \right| \leq \varepsilon \|f\|_2^2 \tag{4}$$

if $\Lambda_{j'}$, $\Lambda_{j''}$ are N-separated for any $j' \neq j''$.

Proof. There is a $\delta > 0$ such that $\int_e |f|^2 \leq (\varepsilon/4)\|f\|_2^2$ for any $e \subset \mathbb{T}$ satisfying $|e| \leq \delta$ and for any $f \in L^2(\Lambda_0)$. Consider the N-th Fejer polynomial p_N of χ_E (=the characteristic function of E). Put $r_N := |\chi_E - p_N|$, $A_1 := \{r_N > \varepsilon/4\}$, $A_2 := \{r_N \leq \varepsilon/4\}$. Clearly, $0 \leq p_N \leq 1$, $0 \leq r_N \leq 1$ and $|A_1| < \delta$ if $N = N(\varepsilon, \delta)$ is sufficiently large. Now,

$$\begin{aligned} \left| \int_E |f|^2 - \int_{\mathbb{T}} |f|^2 p_N \right| &\leq \int_{\mathbb{T}} |f|^2 r_N = \int_{A_1} + \int_{A_2} \leq \\ &\leq (\varepsilon/4)\|f\|_2^2 + (\varepsilon/4)\|f\|_2^2 = (\varepsilon/2)\|f\|_2^2 \end{aligned}$$

if $f \in L^2(\Lambda_0)$. This estimate holds, in particular, if we replace f by f_j. On the other hand, $\Lambda_{j'}$ and $\Lambda_{j''}$ being N-separated, $\int_{\mathbb{T}} f_{j'} \overline{f_{j''}} p_N = 0$ for $j' \neq j''$ whence

$$\int_{\mathbb{T}} |f|^2 p_N = \sum_j \int_{\mathbb{T}} |f_j|^2 p_N.$$

Thus $\triangle \leq (\varepsilon/2)\|f\|_2^2 + (\varepsilon/2)\sum_j \|f_j\|_2^2 = \varepsilon \|f\|_2^2$. □

B) The almost orthogonal splitting (in $L^2(E)$) of any function $f \in L^2(\Lambda_0)$ can be achieved in a canonical way. *For any $\varepsilon > 0$ there are a positive number M $(= M(\varepsilon))$ and a sequence (Λ_j) of subsets of Λ_0 such that $\Lambda_{j'} \cap \Lambda_{j''} = \emptyset$ $(j' \neq j'')$, $\operatorname{diam} \Lambda_j \leq M$, and $\triangle \leq \varepsilon \|f\|_2^2$ for any $f \in L^2(\Lambda_0)$ where $f_j := \sum_{k \in \Lambda_j} \widehat{f}(k) z^k$, and $\triangle$ is defined in* (4).

Proof. According to 3D) and 3E), there is a $\kappa \in (0,1)$ such that no more than $CM^{1-\kappa}$ of any M successive integers are in Λ_0. Hence any interval of length $M \in \mathbb{N}$ contains a subinterval of length cM^κ free of Λ_0. Delete from $[pM, (p+1)M]$ $(p \in \mathbb{Z})$ an interval J_p of length $\geq cM^\kappa$ free of Λ_0. Then $\Lambda_0 \subset \mathbb{R} \backslash \cup J_p = \cup I_j$ where I_j are intervals and $\operatorname{dist}(I_{j'}, I_{j''}) \geq cM^\kappa$ $(j' \neq j'')$. Putting $\Lambda_j := \Lambda_0 \cap I_j$ we get the desired result (note that $cM^\kappa \to +\infty$ as $M \to +\infty$, and we may apply A)). □

6. Rare Spectra. We call a set $\Lambda \subset \mathbb{Z}$ *N-rare* if $|\ell' - \ell''| > N$ for any $\ell', \ell'' \in \Lambda$, $\ell' \neq \ell''$.

Suppose $\Lambda \subset \Lambda_0$ is γ-sufficient for E $(\gamma > 0)$. Then there are $M = M(\gamma) \in$ $\in \mathbb{N}$ and $\gamma' = \gamma'(\gamma) > 0$ such that $\Lambda \cup \Lambda^*$ is γ'-sufficient for E where Λ^* is an arbitrary M-rare subset of Λ_0.

Proof. By virtue of Sect. 2, there is a $\gamma^* = \gamma^*(\gamma) > 0$ such that all sets $\Lambda \cup \{\ell\}$ with $\ell \in \Lambda_0$ are γ^*-sufficient for E. Put $M := N(\gamma^*/2)$ and consider the subdivision (Λ_j) of Λ_0 described in 5B) and corresponding to $\varepsilon := \gamma^*/2$:

$$\Lambda_0 = \cup \Lambda_j, \quad \Lambda_{j'} \cap \Lambda_{j''} = \emptyset \quad (j' \neq j'').$$

If $f \in L^2(\Lambda \cup \Lambda^*)$, then $f = \sum f_j$, $f_j \in L^2((\Lambda \cup \Lambda^*) \cap \Lambda_j)$, $\int_E |f|^2 \geq$ $\geq \sum_j \int_E |f_j|^2 - (\gamma^*/2)\|f\|_2^2$. But since Λ^* is M-rare, $\Lambda^* \cap \Lambda_j$ is either empty or a singleton. Hence $\operatorname{spec} f_j \subset \Lambda$ with a possible exception of a single element. But then (according to our choice of γ^*) $\int_E |f_j|^2 \geq \gamma^* \|f_j\|_2^2$, and $\int_E |f|^2 \geq$ $\geq \gamma^* \sum \|f_j\|_2^2 - (\gamma^*/2)\|f\|_2^2 = (\gamma^*/2)\|f\|_2^2$. Thus we may put $\gamma' := \gamma^*/2$. □

7. The End of the Proof of the Mikheyev Theorem.

A) A set $\Lambda^* \subset \Lambda_0$ is called *negligible* if $(E', \Lambda \cup \Lambda^*)$ is a strong a-pair whenever (E', Λ) is $(\Lambda \subset \Lambda_0)$. If Λ^* is negligible, then (E', Λ^*) is a strong a-pair, and a finite union of negligible sets is negligible. We have shown in Sect.2 that any finite subset of Λ_0 is negligible.

B) The result obtained in Sect.6 can be stated as follows: if $\Lambda \subset \Lambda_0$ and (E', Λ) is a strong a-pair, then so is $(E', \Lambda \cup \Lambda^*)$ whenever $\Lambda^* \subset \Lambda_0$ is an M-rare set $(M = M(\Lambda))$. This remark yields a useful criterion of negligibility after we introduce the following notion. We say that a set $\Lambda \subset \Lambda_0$ belongs to the class $\mathcal{M}_1$ if for any $M > 0$ there is a finite part $\Lambda(M)$ of Λ such that $\Lambda \setminus \Lambda(M)$ is M-rare.

All elements of $\mathcal{M}_1$ are negligible.

Proof. Suppose $\Lambda \subset \Lambda_0$, and (E', Λ) is a strong a-pair. Then Λ is γ-sufficient for E (with a positive γ). Any $\Lambda^* \in \mathcal{M}_1$ can be represented as $\Lambda_1^* \cup \Lambda_2^*$ where Λ_1^* is $M(\gamma)$-rare and Λ_2^* is finite. But then $\Lambda \cup \Lambda_1^*$ is $\widehat{\gamma}$-sufficient for E (with a $\widehat{\gamma} > 0$) by Sect. 6. Hence $(\Lambda \cup \Lambda_1^*) \cup \Lambda_2^*$ is also $\widehat{\widehat{\gamma}}$-sufficient for E (with a $\widehat{\widehat{\gamma}} > 0$), since Λ_2^* is negligible. □

C) Here we define an increasing sequence $(\mathcal{M}_j)_{j \in \mathbb{N}}$ of classes of subsets of Λ_0. We say that a set $\Lambda \subset \Lambda_0$ belongs to $\mathcal{M}_{j+1}$ if for any $M > 0$

$$\Lambda = \Lambda_1 \cup \bigcup_{k=2}^{N} \Lambda_k$$

where Λ_1 is M-rare, and $\Lambda_k \in \mathcal{M}_j$, $k = 2, \dots, N$. It is natural to define $\mathcal{M}_0$ as the set of all singletons in Λ_0. Repeating the argument from B) we conclude that $\cup \mathcal{M}_j$ *consists of negligible sets.*

D) *If $\Lambda \subset \Lambda_0$ contains no cluster of length ℓ, then $\Lambda \in \mathcal{M}_{\ell-1}$* ($\ell \in \mathbb{N}$).

Proof. Our assertion being obvious for $\ell = 1$, suppose it to be true for an $\ell \geq 2$. Take a set $\Lambda^* \subset \Lambda_0$ containing no cluster of length $\ell + 1$. Put $\Lambda_j^* := \{\, k \in \Lambda^* : k + j \in \Lambda^* \,\}$. Then for any $M \in \mathbb{N}$

$$\Lambda^* = \bigcup_{j=1}^{M} \Lambda_j^* \cup R, \quad R := \Lambda^* \setminus \bigcup_{j=1}^{M} \Lambda_j^*,$$

R is M-rare, and Λ_j^* contains no cluster of length ℓ. Hence $\Lambda_j^* \in \mathcal{M}_{\ell-1}$, and $\Lambda^* \in \mathcal{M}_\ell$. □

E) Let us finish the proof. The lengths of clusters contained in Λ_0 are bounded (see 3D)). Hence $\Lambda_0 \in \mathcal{M}_\ell$ for an ℓ (by D)). But then Λ_0 is negligible (by C)) whence (E', Λ_0) is a strong a-pair. □

8. Lacunary Sets. A set $A \subset \mathbb{Z}$ is called *Q-thin* if it is contained in $\mathbb{N}$ or in $-\mathbb{N}$ and if

$$\inf\{\, |n|/|m| : n, m \in A,\ |n| > |m| \,\} \geq Q.$$

A set Λ of integers is called *lacunary* (in the sense of Hadamard) if there are $N \in \mathbb{N}$ and $Q > 1$ such that $\Lambda \cap [N, +\infty)$ and $\Lambda \cap (-\infty, -N]$ are Q-thin. In this section we prove that any lacunary sets belongs to $\Lambda(4)$. This fact implies the sparseness of any lacunary set (by the Mikheyev theorem).

A) Suppose $r \in \mathbb{Z}$, $r \neq 0$, $\Lambda \subset \mathbb{Z}$. Denote by $\rho_\Lambda(r)$ the number of representations of r as the difference of elements of Λ. To be more precise, put $\mathcal{V}_\Lambda(r) := \{\, (p, q) \in \Lambda \times \Lambda : p - q = r \,\}$, $\rho_\Lambda(r) := \operatorname{card} \mathcal{V}_\Lambda(r)$.

Denote by Λ^4 the set of all quadruples $\ell = (\ell_1, \ell_2, \ell_3, \ell_4)$ where $\ell_j \in \Lambda$. Put $\Lambda_0^4 := \{\, \ell \in \Lambda^4 : \sum_{j=1}^{4} \ell_j = 0 \,\}$. If $f \in \mathcal{P}(\Lambda)$, then

$$\begin{aligned}
\|f\|_4^4 &= \int_{\mathbb{T}} \sum_{\ell \in \Lambda^4} \widehat{f}(\ell_1)\widehat{f}(\ell_2)\overline{\widehat{f}(\ell_3)\widehat{f}(\ell_4)} z^{\ell_1+\ell_2-\ell_3-\ell_4}\, dm(z) = \\
&= \sum_{\ell \in \Lambda_0^4} \widehat{f}(\ell_1)\widehat{f}(\ell_2)\overline{\widehat{f}(\ell_3)\widehat{f}(\ell_4)} \le \\
&\le \Bigg(\sum_{\ell_1=\ell_3,\, \ell_2=\ell_4} + \sum_{n \in \mathbb{Z}\setminus\{0\}} \ \sum_{\ell_1-\ell_3=\ell_2-\ell_4=n} \Bigg) |\widehat{f}(\ell_1)\widehat{f}(\ell_2)\widehat{f}(\ell_3)\widehat{f}(\ell_4)| = \\
&= \sum_{(\ell_1,\ell_2)\in \Lambda\times\Lambda} |\widehat{f}(\ell_1)|^2 |\widehat{f}(\ell_2)|^2 + \sum_{n\in\mathbb{Z}\setminus\{0\}} \Bigg(\sum_{\ell\in\Lambda,\, \ell+n\in\Lambda} |\widehat{f}(\ell)|\,|\widehat{f}(\ell+n)| \Bigg)^2 \le \\
&\le \Bigg(\sum_{\ell\in\Lambda} |\widehat{f}(\ell)|^2 \Bigg)^2 + \sum_{n\in\mathbb{Z}\setminus\{0\}} \rho_\Lambda(n) \sum_{\ell,\, \ell+n\in\Lambda} |\widehat{f}(\ell)|^2 |\widehat{f}(\ell+n)|^2 \le \\
&\le \Big(\sup \rho_\Lambda + 1 \Big) \|f\|_2^4.
\end{aligned}$$

This estimate shows that

$$\sup \rho_\Lambda < +\infty \implies \Lambda \in \Lambda^4 . \tag{5}$$

B) Now we have to prove that ρ_Λ is bounded if Λ is lacunary. We start with the following remark. *The number of elements of a Q-thin set contained in $[a, b]$ where $Q > 1$ and $a > 0$ does not exceed* $\log_Q(b/a) + 1$.

Proof. Suppose $a \le a_1 < a_2 \ldots < a_m \le b$ and a_j belongs to a Q-thin set. Then $a_{k+1} \ge Qa_k$ $(k = 1, \ldots, m-1)$, and $b \ge a_m \ge Q^{m-1}a_1 \ge Q^{m-1}a$. □

C) Suppose Λ is lacunary, $r \in \mathbb{N}$. Clearly,

$$\mathcal{V}_\Lambda(r) = \mathcal{V}' \cup \mathcal{V}'' \cup \mathcal{V}'''$$

where $\mathcal{V}' := \{(p,q) \in \mathcal{V}_\Lambda(r)\colon pq < 0,\ p \ge |q|\}$, $\mathcal{V}'' := \{(p,q) \in \mathcal{V}_\Lambda(r)\colon pq < 0,\ p \le |q|\}$, $\mathcal{V}''' := \{(p,q) \in \mathcal{V}_\Lambda(r) : pq > 0\}$. If $(p,q) \in \mathcal{V}'$, then $p > 0$, $q < 0$, $r = p + |q|$, $p \le r \le 2p$, $r/2 \le p \le r$. By virtue of B), the number of points $p \in \Lambda$ satisfying the last estimate is not greater than $N + \log_Q 2 =: \mu(\Lambda)$ (N and Q are the lacunarity parameters mentioned at the beginning of this section). The number of elements $(p,q) \in \mathcal{V}'$ with the given p does not exceed p. Hence $\operatorname{card} \mathcal{V}' \le (\mu(\Lambda))^2$. The same argument shows that $\operatorname{card} \mathcal{V}'' \le (\mu(\Lambda))^2$. If $(p,q) \in \mathcal{V}'''$, then $p, q \in \mathbb{N}$, $p > q$; hence $q \le Q^{-1}p$, $p \ge r \ge (1 - Q^{-1})p$, $r \le p \le (Q/(Q-1))r$. The number of such p does not exceed $N + 2 - \log_Q(Q-1) =: \ell(\Lambda)$, and $\operatorname{card} \mathcal{V}''' \le (\ell(\Lambda))^2$. Thus ρ_Λ is bounded on $\mathbb{N}$. The estimate of $\rho_\Lambda(-r)$ (for $r \in \mathbb{N}$) is analogous. □

Combining this result with (5) and with the Mikheyev theorem, we see that *any finite union of lacunary sets is sparse.*

9. Concluding Remarks.

A) The sparseness of a lacunary set can be given a direct proof. We sketch it in A) – D). Suppose $f \in \mathcal{P}(\Sigma), \Sigma \subset \mathbb{Z},\ S \subset \mathbb{T}$. Then

$$\int_S |f|^2 = \sum_{(p,q)\in\Sigma\times\Sigma} \widehat{f}(p)\overline{\widehat{f}(q)}\,\overline{\widehat{\chi}_S(p-q)} = \sum_{p\in\Sigma} |\widehat{f}(p)|^2|S| + k,$$

$$k := \sum_{(p,q)\in\Sigma\times\Sigma,\, p\neq q} \widehat{f}(p)\overline{\widehat{f}(q)}\,\overline{\widehat{\chi}_S(p-q)} \leq \left(\sum_{(p,q)\in\Sigma\times\Sigma} |\widehat{f}(p)\widehat{f}(q)|^2\right)^{1/2} \sigma^{1/2} =$$
$$= \|f\|_2^2\sigma^{1/2}$$

where $\sigma = \sum_{p,q\in\Sigma,\, p\neq q} |\chi_S(p-q)|^2 = \sum_{r\in\mathbb{Z}\setminus\{0\}} \rho_\Sigma(r)|\widehat{\chi}_S(r)|^2$. Thus

$$\int_S |f|^2 \leq c\|f\|_2^2, \quad c := |S| + \sigma^{1/2}, \tag{6}$$

and $(P_S, \widehat{P}_\Sigma)$ *is a strong a-pair whenever* $c < 1$ $(P_S f := \chi_S f,\ \widehat{P}_\Sigma f := \sum_{k\in\Sigma} \widehat{f}(k)z^k)$.

B) Put $\Sigma_N := \Sigma\setminus[-N, N]$. The above estimate shows that *if* $|S| < 1$, $\sigma < +\infty$ *and* $N = N(S,\Sigma)$ *is sufficiently large, then* $(P_S, \widehat{P}_{\Sigma_N})$ *is a strong a-pair*. The rest follows the scheme similar to Sect. 2.2–2.4. *If* $|S| < 1$, $\sigma < +\infty$, *then* $\dim\mathcal{E}(S,\Sigma) < +\infty$.

Proof. If $f \in \mathcal{E}(S,\Sigma)$, then $f = P_S\widehat{P}_\Sigma f = P_S\widehat{P}_{\Sigma'_N} f + P_S\widehat{P}_{\Sigma_N} f$ where $\Sigma'_N = \Sigma \cap [-N, N]$. Put $R_N := I - P_S\widehat{P}_{\Sigma_N}$; clearly, $R_N f = P_S\widehat{P}_{\Sigma'_N} f$. For N sufficiently large R_N is invertible, and $f = R_N^{-1}(P_S\widehat{P}_{\Sigma'_N})f =: Tf$. But the range of $\widehat{P}_{\Sigma'_N}$ is of finite dimension, and so is T. □

C) Suppose $S_0 \subset S \subset \mathbb{T}$, $0 < |S_0|, |S| < 1$. *For any* $\varepsilon > 0$ *there is a* $\zeta \in \mathbb{T}$ *such that* $|S| < |S \cup \zeta S_0| < |S| + \varepsilon$. This fact (an easy analog of Sect.2.2) implies the following corollary. *If* $|S| < 1$, $\Sigma \subset \mathbb{Z}$, *then either* $\mathcal{E}(S,\Sigma) = \{0\}$ *or for any* $\varepsilon > 0$ *there is a set* $S^* \subset \mathbb{T}$ *such that* $S \subset S^*$, $|S^* \setminus S| < \varepsilon$ *and* $\dim\mathcal{E}(S^*,\Sigma) = +\infty$ (see Sect.2.3).

D) Combining B) and C) we conclude that if

$$|S| < 1, \quad \sigma < +\infty, \tag{7}$$

then $(P_S, \widehat{P}_\Sigma)$ is an a-pair. But $\widehat{P}_\Sigma = \widehat{P}_{\Sigma_N} + \widehat{P}_{\Sigma'_N}$, $I - P_S\widehat{P}_{\Sigma_N}$ is invertible for large N, and $\dim\widehat{P}_{\Sigma'_N}L^2 < +\infty$. Thus $(P_S, \widehat{P}_\Sigma)$ is a strong a-pair if (7) is fulfilled (see 1.2B)). It remains to note that $\sigma < +\infty$ if Σ is lacunary (see 8B)).

E) Functions with sparse spectra obey the "all or nothing" principle: they are either everywhere (i.e. their support is $\mathbb{T}$) or nowhere (i.e. vanish identically). Smoothness properties of functions f with sparse spectra are of the same kind: if f is not everywhere smooth, then it is smooth nowhere. The formal statement is this: *if* $f \in L^2(\mathbb{T})$, $\operatorname{spec} f$ *is sparse and* f *coincides with a* $C^{n+1}(\mathbb{T})$*-function on a set of positive length, then* $f \in C^n(\mathbb{T})$. The proof for lacunary spectra is in Zygmund [1]. It can be adapted to sparse spectra and to the situation of the Amrein–Berthier theorem (if $f \in L^2(\mathbb{R})$, $|\operatorname{spec} f| < +\infty$ and f coincides with a smooth function outside a set of finite length, then f is smooth).

F) Let us use a modification of (7) to obtain a variant of the Amrein–Berthier theorem. *Suppose* $S \subset \mathbb{R}$, $|S| < +\infty$, $b > 0$, $\Lambda \subset \mathbb{Z}$ *is lacunary*, $\Sigma := \bigcup_{\lambda\in\Lambda} [\lambda - b, \lambda + b]$. *Then* (S, Σ) *is a strong a-pair* (though $|\Sigma| = +\infty$).

Proof. Take a large $N > 0$ and put $\Lambda_N := \{\lambda \in \Lambda : |\lambda| > N\}$, $\Sigma_N := \bigcup_{|\lambda|>N} I_\lambda$, $I_\lambda := [\lambda - b, \lambda + b]$. If ess spec $f \subset \Sigma_N$, then $f = \sum_{\lambda\in\Lambda_N} f_\lambda$, ess spec $f_\lambda \subset I_\lambda$ (we may assume that I_λ with $|\lambda| > N$ are disjoint). As in A) we get

$$\int_S |f|^2 = \mathrm{I} + \mathrm{II}, \quad \mathrm{I} := \sum_{\lambda\in\Lambda_N} \int_S |f_\lambda|^2,$$

$$\mathrm{II} := \sum_{\lambda\neq\mu,\ \lambda,\mu\in\Lambda_N} \iint_{I_\lambda I_\mu} \widehat{f}_\lambda(\xi)\overline{\widehat{f}_\mu(\eta)\,\widehat{\chi}_S(\xi-\eta)}\,d\xi\,d\eta.$$

Clearly, $\|P_S\widehat{P}_{I_\lambda}\| = \|P_S\widehat{P}_{[-b,b]}\| =: q < 1$.
Hence

$$\mathrm{I} \le \sum_{\lambda\in\Lambda_N} \|P_S f_\lambda\|^2 = \sum_{\lambda\in\Lambda_N} \|P_S\widehat{P}_{I_\lambda} f_\lambda\|^2 \le q^2 \sum_{\lambda\in\Lambda_N} \|f_\lambda\|^2 \le q^2\|\widehat{f}\|^2 = q^2\|f\|^2.$$

It is easy to see that

$$\mathrm{II} \le \|f\|^2 \sum_{\lambda\neq\mu,\ \lambda,\mu\in\Lambda_N} \iint_{I_\lambda I_\mu} |\widehat{\chi}_S(\xi-\eta)|^2\,d\xi\,d\eta \le$$

$$\le 2b\|f\|^2 \sum_{\lambda\neq\mu,\ \lambda,\mu\in\Lambda_N} \int_{-2b}^{2b} |\widehat{\chi}_S(\lambda-\mu+x)|^2\,dx.$$

The set Λ being lacunary, we may pick a large N and get the following estimate: $\mathrm{II} \le \sigma\|f\|^2$, $q^2+\sigma < 1$, so that (S, Σ_N) is a strong a-pair. But $P_S\widehat{P}_{\Sigma\setminus\Sigma_N}$ is compact, and we may apply 1.2B). □

§4. Supports Strongly Annihilating any Bounded Spectrum. The Logvinenko–Sereda Theorem

In §§2–3 we described some classes of strong a-pairs. But none of these descriptions was exhaustive. For instance, we know that any support and any spectrum of finite volumes are strongly mutually annihilating. But we cannot *characterize all* supports forming a strong a-pair with *any* spectrum of finite volume (we know from 3.9F) that the volume of such a support can be infinite).

In this paragraph we give a *complete* description of supports forming a strong a-pair with any *bounded* spectrum.